

Lesson 07

Division Algorithm-II (Problem solving by checking the last digit)

A kind of problems can be solved by checking the last digit of an integer and by building the arguments on that. You can familiar with the concept and arguments with following activities and examples..

Activity 01

Without calculating the exact value, find the last digit of each of the followings.

- (i) 5002375^6
- (ii) 7112371^9
- (iii) 700002^9
- (iv) 533719^5
- (v) 1222222^5
- (vi) 734202^{17}
- (vii) 634808^{15}
- (viii) 985634^{105}
- (ix) $34^5 + 23^8$
- (x) $33^{18} - 39^{56}$

Example 01

Exactly find each of the following sets.

- (i) {Last digit of $2019^k : k \in \mathbb{N}$ }, (i.e All possible values of last digit of 2019^k ; where $k \in \mathbb{N}$)
- (ii) {Last digit of $2016^k : k \in \mathbb{N}$ }
- (iii) {Last digit of $2017^k : k \in \mathbb{N}$ }
- (iv) {Last digit of $k^2 : k \in \mathbb{N}$ }
- (v) {Last digit of $k^4 : k \in \mathbb{N}$ }

Answer:-

(i) You can empirically check, last digits of first few terms; $2019, 2019^2, 2019^3, 2019^5 \dots$
Those are, 9, 1, 9, 1 ... respectively and the cycle 9,1 repeats continuously. More precisely,
the last digits of the sequence (2019^k) is coincided with last digits of the sequence (9^k) .
Hence $\{\text{Last digit of } 2019^k : k \in \mathbb{N}\} = \{1, 9\}$.

(ii) The last digits of the sequence (2016^k) is coincided with last digits of the sequence (6^k) .
Hence $\{\text{Last digit of } 2016^k : k \in \mathbb{N}\} = \{6\}$.

(iii) The last digits of the sequence (2017^k) is coincided with last digits of the sequence (7^k) .
Hence $\{\text{Last digit of } 2017^k : k \in \mathbb{N}\} = \{7, 9, 3, 1\}$.

(iv) Let's find first the last digits of first 10 terms; those are the last digits of $1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2$ and 10^2 . Which are 1, 4, 9, 6, 5, 6, 9, 4, 1, and 0 respectively. Also one can easily show that this pattern continued. Hence $\{\text{Last digit of } k^2 : k \in \mathbb{N}\} = \{1, 4, 9, 6, 5, 0\}$.

(v) Let's find first the last digits of first 10 terms; those are the last digits of $1^4, 2^4, 3^4, 4^4, 5^4, 6^4, 7^4, 8^4, 9^4$ and 10^4 . Which are 1, 6, 1, 6, 5, 6, 1, 6, 1, and 0 respectively. Also one can easily show that this pattern continued. Hence $\{\text{Last digit of } k^4 : k \in \mathbb{N}\} = \{1, 5, 6, 0\}$.

Activity 02



(i) Prove that last digit of sum of any two fourth powers of integers cannot be 3. (Hint: consider all possible pair wise sum of the exact values of the set in part (v) of above example)

(ii) Prove that last digit of sum of any three fourth powers of integers cannot be 4.

(iii) Can you state a similar problem for sum of four fourth powers of integers? If can, do it right now. If can't justify.

In mathematics same arguments (or concepts) can be used to solve different problems. On the other hand, a problem can be transformed in to different versions. For example consider the following two variations of part (i) of Activity 02. Did you notice that same argument of answer to the part (i) of Activity 3, can be used to answer the two versions? At the same time one can use completely different arguments to prove two versions.

- Prove that there are no integer solutions for the equation $a^4 + b^4 = 2023$
- Prove that there are infinitely many positive integers which cannot be written as sum of two fourth powers.

Did you notice that, we didn't refer the division algorithm introduced in Lesson 03? However in a hidden way, we used the division algorithm, more precisely the following; "Any integer can be written as $10k + r$, where $0 \leq r < 10$ ". Since we are using the decimal system (base 10), automatically the last digit of any integer is equal to the remainder when it is divided by 10. As some examples, by directly using the division algorithm, we can reformulate the part (i) of example 3.3.5 and part (v) of activity 3 as follows;

- Find all possible remainders when dividing 2019^k by 10, where $n \in \mathbb{N}$.
- Prove that the sum of any two fourth powers of integers cannot be of the form $10k + 3$.

Generally, we can use the same techniques and arguments not only for the $10k + r$ forms, but for the $ak + r$ forms, where $0 \leq r < q$. You practiced some in Lesson 03. You will be practicing more, through following examples and activities.

Example 02

Integers of the form $4k + 3$ never are written as the sum of two squares.

Answer:

By division algorithm, any integer can be written as $4m$, $4m + 1$, $4m + 2$ or $4m + 3$.

Now $(4m)^2 = 4 \cdot 4m^2$, $(4m + 1)^2 = 4(4m^2 + 2m) + 1$, $(4m + 2)^2 = 4(4m^2 + 4m + 1)$

Division Algorithm (Second Version)

From the earlier version of Division Algorithm, we can derive a new version of Division Algorithm. The two versions are equivalent, but this second version has some symmetric appearance in the remainder, that will sometimes shorten the calculations. We prove the second version using the first version of Division Algorithm. Hence the new one becomes corollary to the first version.

Corollary (Division Algorithm, second version)

If a and b are integers with $b > 0$, then there exists unique integers q and r satisfying

$$a = qb + r, \text{ where } -\frac{1}{2}b < r \leq \frac{1}{2}b.$$

Proof:-

By first version of Division Algorithm, $a = q'b + r'$, where $0 \leq r' < b$ and q', r' are unique.

Case (i):- When $0 \leq r' \leq \frac{b}{2}$

Choose $q' = q$, $r' = r$.

Case (ii):- When $\frac{b}{2} < r' < b$

Let $r = r' - b$, and $q = q' + 1$

Observe that then, $a = b(q' + 1) + (r' - b)$ with $-\frac{1}{2}b < (r' - b) < 0$.

Also the uniqueness of q' and r' ensure the uniqueness of q and r . ■

Example 03

- (i) The fourth power of any integer is either of the form $5k$ or $5k + 1$
(ii) An integer of the form $4k + 2$ never a perfect square.

Answer:-

(i) Let n be any integer. Then by division algorithm (second version), n is of the form $5k$, $5k + 1$, or $5k + 2$.. Observe that,

$$(5k \pm r)^4 = (5k)^4 \pm {}^4C_1(5k)^3r + {}^4C_2(5k)^2r^2 \pm {}^4C_3(5k)r^3 + r^4$$

Observe that for $r = 0, \pm 1, \pm 2$, the first four terms of the right hand side of the above identity is divided by 5. Possible values for r^4 is 0, 1, or 16. But $16 = 5 \cdot 3 + 1$. This shows that $(5k + r)^4$ is of the form $5k$ or $5k + 1$ for $r = 0, \pm 1, \pm 2$.

Hence the fourth power of any integer is either of the form $5k$ or $5k + 1$.

(ii) Let n be any integer. By division algorithm(second version), n is of the form $4k$, $4k + 1$, or $4k + 2$.

Now $(4k)^2 = 4 \cdot 4k^2$, $(4k + 1)^2 = 4(4k^2 + 2k) + 1$, and $(4k + 2)^2 = 4(4k^2 + 4k + 1)$.

Hence square of any integer is either of the form $4k$ or $4k + 1$. Therefore an integer of the form $4k + 2$ is never a perfect square.